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| Course Code |            |
| Course Name | TypeScript |

|                    |      |                   |                       |
|--------------------|------|-------------------|-----------------------|
| Duration (in days) | 2    | Proficiency Level | Front End Development |
| Pre-requisites     | None | Target Audience   | Campus Hires          |

## Learning Outcome

At the end of the program, participants will be able to learn:

- Understand the fundamentals of TypeScript and its benefits over JavaScript.
- Master TypeScript syntax, data types, and type annotations.
- Utilize TypeScript interfaces and classes for creating reusable and maintainable code.
- Apply TypeScript features such as decorators, generics, and modules.

## Day-wise Session Plan

| Day | Unit       | Objective(s)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Hours |
|-----|------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 1   | TypeScript | <ul style="list-style-type: none"> <li>• What is TypeScript?</li> <li>• Benefits of TypeScript</li> <li>• Setup the Environment</li> <li>• Basic Data Types</li> <li>• Arrays</li> <li>• Tuples</li> <li>• Enum</li> <li>• Any and void</li> <li>• null and undefined</li> <li>• Type Inference</li> <li>• Type Casting</li> <li>• Difference between let and var</li> <li>• Const declaration</li> <li>• Writing and Using Classes</li> <li>• Constructor method</li> <li>• Inheritance of classes</li> <li>• Type Casting</li> <li>• Type Assertion</li> <li>• Abstract class</li> <li>• Interface Declaration and Initialization with an object</li> <li>• When to Use Generic Functions</li> </ul> | 16    |